

Collection of problems for Lectures 1-5

Below are a collection of 85 problems related to the material Lectures 1 through 5. At the end of Lectures 1, 2, 3 and 4, I will assign a few questions from this set as the group assignment. Answers to the each assignment are due before the start of the next class. We will post solutions for all of these problems as the week progresses. There will be no assignment for the material in Lecture 5, but please study the questions and suggested solutions when preparing for the midterm exam.

These are group assignments. Each group should submit one solution set for each assignment. The name of every group member who participated in the exercise should be printed at the top of the assignment. One person from the group must attest that the names of the participating members are accurate. If possible, that person should sign the document. You may hand write (rather than type) your solutions – whichever is more convenient. Then scan/photograph your solutions and upload them to the canvas course website.

Problem 1 True, False, Uncertain? [Explain your answer.] Suppose that coffee and cream are demand complements and are traded in competitive markets. Suppose also that there was a reduction in the supply of cream due to a reduction in dairy livestock from disease. *The equilibrium price of coffee will decrease.*

Problem 2 The following demand and supply functions describe the market for natural gas. The g subscripts denote “natural gas” and the o subscripts denote oil. The s and d superscripts refer to “supply” and “demand”, respectively.

$$Q_g^s = 14 + 2p_g + 0.25p_o,$$

$$Q_g^d = 3.75p_o - 5p_g.$$

- (a). Is oil a demand substitute or demand complement to natural gas? How do you know?
- (b). Calculate the equilibrium price of natural gas if the price of oil is fixed at $p_o = 8$.
- (c). Calculate the (point) elasticity of supply for natural gas at the equilibrium price above in (b).
- (d). Suppose for exogenous reasons that the price of oil rises from $p_o = 8$ to $p_o = 16$. What happens to the equilibrium price of natural gas?

Problem 3 The government is interested in the domestic market for wheat. The government’s chief economist estimates the following equations for the demand and supply functions:

$$Q^d = 1500 - 100p,$$

$$Q^s = 500 + 300p.$$

Quantities are measured in thousands of bushels; prices measured in dollars per bushel.

- (a). Calculate the equilibrium price and quantity.
- (b). Calculate the (point) elasticities of supply and demand at the equilibrium values in part (a).
- (c). The government is proposing a \$3/bushel price-support program. What this means is that the government will buy up wheat as necessary to guarantee that the market price will be \$ 3 per bushel. How much wheat will the government need to purchase under this program?

Problem 4 Suppose that there are two kinds of firms that sell plumbing services to a market. The firms are price takers; there is a single market price for an hour of plumbing service. Firms can decide how much to supply at that price. The first type of firm (type “A”) can sell up to 1000 hours of service per week at an incremental (i.e., marginal) cost of \$20 per hour for each hour supplied (this \$20 represents all of the firms additional costs – e.g., the cost of insurance, wages, etc.); beyond 1000 hours of service, it can sell an additional 400 hours of service per week with each additional hour costing \$30 (the higher cost represents the fact that the firm will have to pay overtime wages to its workers). It cannot supply more than 1400 hours per week. The second type of firm, type “B”, is similar to type “A”, except that it has different costs and capacities. It can supply up to 500 hours of service per week at a cost of \$25 per hour; it can supply an additional 200 hours per week for a cost of \$35 per hour; it cannot supply more than 700 hours per week.

Hint: The unit costs given above should be understood to represent each firm’s additional cost of supplying that unit. As such, these costs provide the critical market price at which the firm would be willing to supply service. For example, at a market price of $p = 21$, type “A” firms would supply 1000 and make a “profit” or “producer surplus” of $(21 - 20)(1000) = 1000$ for that supply; type “B” firms would be unwilling to supply at this price since it exceeds its costs of supplying.

Suppose that there are 10 firms of each type in this market (a total of 20 firms). Draw the *market* supply curve. Now suppose that the market price of plumbing services is $p^* = \$32/\text{hour}$. What is the total producer surplus in this market at this price?

Problem 5 Suppose there are 300 buyers and 300 sellers in a market for apples. Each seller has exactly one bushel of apples available to sell; each buyer has demand for at most one bushel of apples to buy.

Suppose 200 sellers have a cost of \$10 per bushel and 100 sellers have a cost of \$30 per bushel. These costs represent what the sellers give up by selling a bushel of apples (i.e., this is their opportunity cost of the apples). Suppose 200 of the buyers have a value for apples of \$20 per bushel and 100 buyers have a value of \$40 per bushel. These values represent the most that the buyers are willing to pay.

- (a). What is the equilibrium price in this market and how many bushels of apples are traded? Determine this outcome by drawing the associated supply and demand graphs.
- (b). How much “profit” do the sellers collectively make? (Hint: This is the area between the market price in (a) and the supply curve.)
- (c). How much “consumer surplus” do the buyers obtain? (Hint: This is the area between the demand curve and the market price in (a)).
- (d). Suppose the government taxes the sellers \$5 per bushel sold. What is the new equilibrium (pre-tax) price? (Hint: the tax is equivalent to raising the seller’s cost of a bushel of apples by exactly \$5. You can repeat the analysis in (a) with this modification.) What happens to producer surplus and consumer surplus? Would it matter if the government taxed the buyers instead (i.e., each buyer’s willingness to pay decreased by \$5)?

Problem 6 In an effort to reduce carbon emissions, the US government has undertaken a broad program of incentives and regulation to eliminate all oil consumption. It anticipates the program will reduce oil consumption from 20 million barrels of oil per day to zero barrels per day, and it anticipates doing this without any substitution into carbon-emitting energy sources.

Suppose that the US estimates are correct and that the elimination of all oil consumption by the US will be accomplished. Does this imply that the US policy will reduce the world’s consumption of oil by 20 million barrels of oil per day? Explain what determines the amount of the world reduction.

Problem 7 An extremely wealthy environmentalist has decided that he wants save rain forests from lumber mills and deforestation. To this end, he buys 100,000 acres of dense rain forest. The agreement gives him exclusive rights to the resources of the forest for this time period. Normally, a purchaser of such a lease would strip the land of its lumber. Rather than logging the forest for valuable mahogany, however, the billionaire chooses to make the forests into nature preserves.

Given his forest purchase includes approximately 1 million mahogany trees (or 100 million cubic feet of wood), he concludes that he has reduced the world’s consumption of rainforest by 100,000 acres and rain-forest mahogany by 100 million cubic feet.

Has the billionaire actually reduced the world’s consumption of rain forests and mahogany by these amounts? What does the conclusion depend upon?

I apologize for the confusing solution to this question. Let me try again here.

There are two ways to solve the problem, each is equally valid.

Problem 8 The following demand and supply curves describe the market for wheat. The w subscript denotes “wheat,” the sb subscript denotes “soybeans” and the c subscript denotes

“corn”. The s and d superscripts denote “supply” and “demand”, respectively.

$$Q_w^s = 40 + p_w,$$

$$Q_w^d = 4p_{sb} + 7p_c - 10p_w.$$

- (a). Are soybeans a substitute or a complement for wheat? How do you know?
- (b). Without any mathematical calculations, describe how the demand curve for wheat shifts as the price of corn rises and everything else remains unchanged.
- (c). What is the equilibrium price of wheat when $p_{sb} = 50$ and $p_c = 40$? What is the equilibrium quantity of wheat traded?
- (d). At the prices above in (c), what is the point elasticity of demand for wheat with respect to the price of wheat?

Problem 9 If the supply and demand curves are given by $Q_d = 30 - p$ and $Q_s = 5 + 4p$, compute the point demand and supply elasticities at equilibrium.

Problem 10 True, False, or Uncertain. [Explain your answer!] *If the demand for Intel-based personal computers is inelastic, then the demand for the Gateway brand of computers (Gateway is a specific brand of Intel-based personal computers) is also generally inelastic.*

Problem 11 Suppose that demand for beer is given by

$$Q_{beer}^d = 120 - 4p_{beer} + 3p_{wine} + 2I,$$

and supply is given by

$$Q_{beer}^s = 100 + 6p_{beer}.$$

- (a). Suppose that $p_{wine} = 20$ and $I = 10$. What is the equilibrium price and quantity of beer?
- (b). Income elasticity:
 - (i). What is the income (point) elasticity of beer at this equilibrium price?
 - (ii). Is beer an inferior good or a normal?
 - (iii). Is it a luxury good?

Problem 12 Consider a market in which the supply curve for memory chips has an elasticity of $\varepsilon_s = 2.0$ and the demand curve has an elasticity of $\varepsilon_d = -0.5$.

- (a). Following an isolated earthquake, the production facilities of several manufacturers are destroyed; no negative effects, however, occur on the demand side of this market. If total output sold falls 10%, approximately how much did the price rise (in percentage terms)?

Explain your answer. [You may assume the given elasticities are constant over this change in output.]

(b). Does the earthquake described in (a) increase or decrease market revenues? Explain.

(c). Suppose that over time, the damaged production facilities are repaired and supply shifts back to precisely its original position. Unfortunately, demand for chips decreases due to reduced demand for computer equipment. Does revenue increase or decrease relative to the original (pre-earthquake) equilibrium? Explain.

Problem 13 True, False, Uncertain? [Explain your answer.] Currently, the supply and demand curves for cigarettes intersect at a price of \$2. Suppose that the elasticity of demand is $\varepsilon = -2.0$. *If the quantity demanded strictly decreases at every price due to a change in tastes, then the equilibrium price will decrease and revenue will increase.*

Problem 14 True, False, Uncertain? [Explain your answer.] Suppose the demand elasticity for oranges is elastic; specifically, suppose that $\varepsilon_d = -2$. *When the market supply for oranges decreases due to bad weather, consumer surplus in the market decreases.*

Problem 15 If the demand elasticity is larger than 1 in absolute value and the supply curve shifts right, do total market expenditures increase or decrease? Explain.

Problem 16 True, False, Uncertain? [Explain your answer.] The July 20, 1999 edition of the *Wall Street Journal* states:

Philip Morris Inc., the company's U.S. tobacco unit, reported operating income of \$1.2 billion, down 3.9% from last year, with lower volume offset by higher prices. The unit's revenue rose to \$4.74 billion from \$3.73 billion, but total volume fell to 51.8 billion cigarettes shipped, down 9.6% from last year.

Therefore, the demand for Cigarettes is elastic.

Problem 17 The demand for haircuts is

$$Q^d = 10,000 - 200p,$$

while the supply of haircuts is

$$Q^s = 1200p - 18,000,$$

assuming that $p \geq 15$; $Q^s = 0$ if $p < 15$.

(a). What is the equilibrium price of haircuts in this market?

(b). What is the own-price elasticity of demand in equilibrium? Is demand elastic or inelastic?

- (c). Compute the consumer and producer surpluses at the equilibrium price in (a).
- (d). Now suppose that supply decreases slightly to

$$Q^s = 800p - 12,000,$$

if $p \geq 15$ and $Q^s = 0$ if $p < 15$. The demand relationship remains unchanged. What is the new equilibrium price? Is demand elastic or inelastic?

- (e). Are market revenues larger at the new price in (d) or at the original price in (a)? Could you have answered this question by using your observations about the elasticities of demand? Explain.

Problem 18 Draw indifference curves reflecting the following statements:

- (a). Advertising statements to the contrary, Coca-Cola and Pepsi-Cola are all the same to me. I don't care which one I get.
- (b). I never eat a Big Mac without a Coca-Cola and I never drink a Coca-Cola without a Big Mac.
- (c). I like apples and oranges. But if you take away half my oranges and double the number of apples I have (and vice versa), I'm just as well off.

Be sure your graphs are well labeled and clearly explained.

Problem 19 True, False, or Uncertain. [Explain your answer!] Suppose a consumer's utility curve is given by $U(x, y) = 2x + 3y + 7$. *Here the goods x and y exhibit diminishing marginal rates of substitution.*

Problem 20 A consumer has income of \$200. Good x costs \$4 per unit and good y costs \$10 per unit.

- (a). Graph the consumer's budget constraint.
- (b). Show what happens to the budget constraint if the price of good y falls to \$5.

Problem 21 You are managing a city budget in which monies are spent on law enforcement and other services (e.g., schools, etc.). Suppose that the city has 5 million Euros of its own funds to allocate between law enforcement and all other services.

You are informed that you are about to receive aid from the federal government to support a special anti-drug law enforcement program. Two programs which might be available are (1) a 1,000,000 Euro grant which must be spent on law enforcement; (2) a matching grant program, in which each euro of local spending on law enforcement is matched by a euro of

federal money which must be spent on law enforcement. The federal matching program limits its payment to each city to a maximum of 1,000,000 Euros.

- (a) Draw the budget constraint relevant for each program.
- (b) Which program would you (the manager) choose if you wish to maximize the satisfaction of the citizens in your jurisdiction?

Problem 22 True, False, Uncertain? [Explain your answer!] If your marginal utility of pizza is larger than your marginal utility of beer, *you should purchase more pizza and less beer.*

Problem 23 True, False, or Uncertain. [Explain your answer!] If Al's income and the prices of every good in the economy simultaneously increase by 10 percent, with the sole exception of the price of beer which remains unchanged, *then Al's consumption of beer must increase.*

Problem 24 True, False, or Uncertain? [Explain your answer!] Robert's marginal utility from consuming chocolate for dessert is 7 and his marginal utility from consuming strawberries for dessert is 4. The price of a dessert of strawberries is £2 and the price of a dessert of chocolate is £3. *Robert should consume chocolate for dessert.* [You may assume that Robert can only consume one or the other for dessert.]

Problem 25 True, False, Uncertain? [Explain your answer.] Suppose that a consumer's indifference curves are convex and that the consumer has optimally allocated his food budget. At the present levels of consumption, the consumer's marginal utility of consuming "pasta" is 9 utils/package and the marginal utility of consuming "steak" is 15 utils/lb. *If the price of steak is currently 5 dollars per pound, then the price of pasta must be 2.50 per package.*

Problem 26 Suppose that Diane's marginal rate of substitution between wine and beer always equals

$$MRS_{\text{wine,beer}} = 3;$$

that is, she would give up three bottles of beer for one bottle of wine at any consumption bundle of beer and wine.

- (a). What can we say about the relationship between these two goods?
- (b). If the price of beer is $p_b = 2$ and the price of wine is $p_w = 7$, what ratio of wine and beer should Diane consume?

Problem 27 (10 points) True, False, or Uncertain. [Explain your answer!] In a two-good world (goods x and y), if the prices of good x and y increase by 5 percent and the income of the consumer decreases by 5 percent, *the marginal rate of substitution of the customer between goods x and y will increase at the optimum.*

Problem 28 True, False, or Uncertain. [Explain your answer!] A consumer's consumption consists of only two goods, good A and good B. When the price of good A rises, the consumer's total expenditures on good B rises. *The consumer's demand for good A is price inelastic.*

Problem 29 True, False or Uncertain? [Explain] The price of Australian wine is \$10 per bottle and the price of French wine is \$25 per bottle. Dominique's marginal utility from her last bottle of Australian wine is 120 utils, while her marginal utility from the last bottle of French wine is 300 utils. *She has rationally allocated her budget across Australian and French wine.*

Problem 30 Consider a simple setting with two goods, x and y . The price of x is $p_x = 20$ and the price of y is $p_y = 15$. Due to various government regulations, consumption is restricted to no more than 2 units of either good (i.e, your consumption choice must satisfy $x \leq 2$ and $y \leq 2$). Consumption is divisible (e.g., it is possible to consumer $x = 1.62$ units). Your income is $I = 60$.

- (a). Graph your constraints for x and y in this context. That is, indicate the set of consumption choices that are feasible (affordable and satisfy the government regulations).
- (b). Suppose that your marginal utility of x is 240 and your marginal utility for y is 200 utils, regardless of the amounts you consume of either good. Exactly how much x and y should you optimally consume given your constraints from part (a).

Problem 31 True, False, Uncertain? [Explain your answer.] *In a two-good world in which both goods are demand substitutes, both goods must have elastic demand curves.*

Problem 32 True, False, Uncertain? [Explain your answer.] Suppose that when the consumer's income rises by 20%, the consumer's consumption of good x only increases 10%. *We can infer that good x is an inferior good.*

Problem 33 A town wants to repair its swimming pool. The annual demand from each of its 5,000 citizens in terms of uses is $q = 18 - 3p$ where p is the price in (dollars) charged for using the pool. If it charges a \$1 admission, it will raise \$75,000 a year, enough to pay for the repairs. An alternative proposal is to charge \$15 a year for a season ticket with no per-usage fees. Which proposal would be better from the point of view of the consumer? Can you quantify the relative advantage of the superior proposal?

Problem 34 When planning this year's budget, Helen Smith decided to purchase 40 compact disks this year at 5 pounds each and to spend the remaining amount of her income on other goods. Before purchasing anything, however, she is offered an opportunity to buy her compact disks from a music club, where she must pay 6 pounds a disk for the first 20 disks but only 4 pounds per disk thereafter. Is she indifferent to the offer, or should she take it? Explain your answer.

Problem 35 The price of apples increases 10 percent, while the price of all other goods (AOG) remains constant. You respond by reducing your consumption of apples by 5 percent.

- (a). Is your demand for apples elastic or inelastic? Explain.
- (b). What happens to total expenditures on apples? Explain.
- (c). Does your consumption on all other goods beside apples increase or decrease? Explain.

Problem 36 Chicago Corp. has its headquarters in Chicago and a branch office in San Francisco. Recently, whenever Chicago Corp. has transferred employees to its San Francisco office, it has raised salaries by 30 percent to take into account the higher cost of housing in the San Francisco Bay area. [You should assume that other goods are roughly the same price as in Chicago; only housing is more expensive.] Nonetheless, while turnover in the Chicago office is only 10 percent per year, the turnover in the San Francisco branch by recently transferred employees is 40 percent. The common complaint by quitting employees is that the cost of living adjustment was insufficient to maintain their standard of living, and hence they preferred to take a job at another company in the Bay area after having settled in.

Taking the transferred workers' complaints seriously, you decide to look carefully at housing prices in the Bay area and adjust their new salaries so as to allow them to purchase the same bundle of goods as if they had remained in Chicago. Your review of the relative housing prices and your resulting calculations suggest that you need to raise their salaries by a full 50 percent rather than merely 30 percent. You are convinced that with this salary adjustment, the transferred workers will be able to purchase the same bundle of goods (housing and other stuff) as they consumed in Chicago.

One year later when you evaluate the annual turnover rates, you notice that while the Chicago turnover rate remains at 10 percent per year, the San Francisco turnover rate has declined to 1 percent implying that you have paid them too much! What was wrong with your cost-of-living-adjustment (COLA) analysis? Draw a graph to illustrate what you have done and what has gone wrong.

Problem 37 (a). Consider a random variable that takes on three possible values, $x_1 = -10$, $x_2 = 15$, $x_3 = 25$, with probabilities $\phi_1 = \frac{1}{2}$ and $\phi_2 = \phi_3 = \frac{1}{4}$. What is the expected value of the random variable?

(b). Consider a second random variable which also takes on three values, $y_1 = -25$, $y_2 = 25$, $y_3 = 45$ with probabilities $\phi_1 = \frac{1}{2}$ and $\phi_2 = \phi_3 = \frac{1}{4}$. What is the expected value of this second random variable?

(c). If a risk averse person were to choose between these two assets, which one would be chosen? [You may assume that the risk averse person has mean-variance preferences.]

Problem 38 Suppose that an Faruk has wealth of \$100,000 and is presented with the following project: with probability $\frac{1}{4}$, the project succeeds and generates profit of \$10,000; with

probability $\frac{1}{2}$, profits are zero; with probability $\frac{1}{4}$, losses are \$10,000. Suppose that the money utility function, $u(x)$, satisfies the following: $u(110,000) = 30$ utils, $u(100,000) = 26$ utils, $u(99,000) = 25.5$ utils and $u(90,000) = 20$ utils.

- (a). What is the expected value (not expected utility) of the project?
- (b). Should Faruk take the project or not?
- (c). If Faruk takes on the project, what is his risk premium?

Problem 39 Suppose that two possible events can occur with a particular project: the Good outcome and the Bad outcome. If the Good outcome occurs, your company *receives* \$400. If the Bad outcome occurs, your company *loses* \$200. Suppose that you (and your company) are risk neutral.

- (a) If the probability of the Good outcome is $\phi_G = .5$, what is the expected value of this project?
- (b) Suppose that you do not know the probability of the Good outcome. What value of ϕ_G would make you indifferent to undertaking this project?
- (c) Now suppose that your company has limited liability for debts equal to your current assets which are \$100. This means that if you take on this project, your downside losses are limited to \$100 because that is all that you must legally pay if the Bad event occurs. Given your limited liability, what is the probability of the Good outcome that would make you indifferent to undertaking this project? That is, answer (b) again, but for the case in which downside losses are limited to \$100.
- (d) At the probability you found in (c), what is the expected value of this project if you had instead \$200 in assets (i.e., you could cover the entire loss of \$ if the Bad outcome arises)?
- (e) Comparing (b) and (c), it appears that for some probability of the good outcome, a risk-neutral company with sufficient assets would reject the project but a risk-neutral company with limited assets and limited liability would accept. It is as if the company is risk loving. Explain why this happened using concepts from class.

Problem 40 True, False, Uncertain? [Explain your answer.] Every additional dollar Joan receives as income provides less additional utility than the previous dollar. *Joan is therefore risk loving.*

Problem 41 For simplicity, suppose there are two kinds of used cars which exist in a 50-50 proportion: good cars worth 5,000,000 pesetas and bad cars (lemons) worth only 1,000,000 pesetas. Also suppose that only the owner truly knows the type of car that it is.

- (a) If both kinds of cars are sold, what is your expected value from buying a used car?

(b) Now suppose that the current owners of good cars (i.e., the sellers) have a reservation value of 4,000,000 pesetas (i.e., they'd rather keep a good quality car than sell it for less than 4,000,000 pesetas); the sellers of bad cars value their cars at only 800,000 pesetas. You know that these reservation prices exist, you just can't distinguish the good car sellers from the bad car sellers. If someone offers you a car at a low price, say 2,500,000, do you buy it? Why or why not?

(c) Is there any market price at which uninformed buyers (i.e., buyers who know everything stated above *but* who do not know whether the car is good or bad by just looking at it) will buy used cars and informed sellers will sell them? What type of cars are sold?

(d) What general problem is this related to?

Problem 42 Elaine has an idea for a small business which is risky. With probability of 0.5, the business will succeed, resulting in profits of \$5,000; with probability of 0.5, the project will be a failure and produce losses of \$2,500. Elaine's initial wealth is \$5,000. The time horizon is sufficiently short that discounting can be ignored.

(a) What is the expected value of the project?

(b) Elaine's utility function for income is approximately $u(x) = \sqrt{x}$. Is she risk averse? Is her marginal utility of income increasing or decreasing?

(c) What is Elaine's expected utility if she undertakes this small business venture? [Hint: Remember that expected utility is calculated by considering the different resulting *wealth* levels which may arise from undertaking a project.] What is Elaine's expected utility if she does nothing and maintains her \$5,000 wealth position with certainty? Should she undertake the project?

(d) What risk premium would Elaine be willing to pay in exchange for receiving the expected value of the project for sure rather than the uncertain gamble?

(e) Suppose that her friend, Kramer, is also thinking about starting a business with identical prospects for success and failure (and identical profits and losses), but the proposed business is completely unrelated to (i.e., independent of) Elaine's project. Elaine and Kramer decide to form a partnership, agreeing to split evenly any gains and/or losses between themselves. Does Elaine's expected utility increase, decrease, or remain the same under the partnership arrangement? Briefly explain.

(f) If Kramer and Elaine weren't such good friends and Elaine couldn't observe how hard Kramer was working, what general problem should Elaine be concerned with? Briefly explain.

(g) Thinking that their partnership is a good idea, Elaine and Kramer decide to place an ad in the paper to get other entrepreneurs with ideas to join their partnership, thereby reducing the risk of their business portfolio. What problem should Elaine and Kramer be worried about, in addition to the problem above in (f)? Briefly explain.

Problem 43 True, False, or Uncertain. [Explain your answer!] The following gamble is offered to a risk-averse person free of charge: win £2000 with probability $1/2$, win £1000 with probability $1/4$, lose £5000 with probability $1/4$. *A risk averse person will not accept this gamble.*

Problem 44 Ellsworth faces a probability of 0.5 that his boss, Gail, might fire him. If he is fired, Ellsworth will remain unemployed for a year, earning nothing. His yearly salary is \$360,000.

- (a) If he prefers earning \$180,000 with a guarantee of employment to his current situation of receiving \$360,000, then what can we say about Ellsworth's preferences toward risk?
- (b) Now suppose that he would be willing to take a pay cut of up to \$200,000 per year (from his current income of \$360,000) in exchange for a guarantee of not being fired; what is his risk premium?
- (c). What might be a reason why employers, like Gail, would choose *not* to offer an employment guarantee such as in (b)?

Problem 45 Suppose that a consumer has the following demand curve:

$$p = 10 - 2q.$$

Because of uncertainty over supply, the market price can be either $p = 6$ (normal supply) or $p = 8$ (shortage of supply). These events occur with equal probability (i.e., $\phi_1 = \phi_2 = \frac{1}{2}$).

- (a). What are the consumer surpluses in each of the two supply states.
- (b). Suppose now that the consumer has to spend \$2 in order to access this market, but that the money must be spent *before* the consumer knows the supply situation. (For example, the consumer must take a day-long journey to get to the market place, and the journey costs him \$2.) What is the expected monetary value of traveling to the market (remember to subtract the costs of travel). Will a risk-neutral consumer wish to travel to this market?
- (c). Now suppose our consumer has preferences over money given by $u(x) = x^2$ and the consumer's wealth is at least 14 (i.e., they can afford the \$2 expense to access the market and have enough money leftover to purchase according to the inverse demand curve $p = 10 - 2q$). Assume that consumer surplus is equivalent to money for this consumer. Will the consumer want to travel to the marketplace? (Note: you may want to plot a few points of $u(x) = x^2$ to understand this person's preferences for risk. Also, you may be able to answer this question without calculating the expected utility of the market trip.)

Problem 46 You manage the marketing department for Vanboard Brokerage Services. The accounting department has advised you that, on average, every new individual account generates an economic profit of \$100 per year, and that the average new customer stays with

Vanboard Brokerage Services for more than 2 years. Your own marketing surveys indicate that if you offer a promotion of a gift certificate worth \$50 to anyone who starts a new customer account at VBS, that you will attract 1000 new customers. What idea from uncertainty theory suggests that this may not be a profitable promotion? Explain.

Problem 47 Suppose that an investor's utility for money is $u(x) = \sqrt{x}$, her wealth is \$1,000,000, and that she is offered an investment opportunity which yields a *loss* of \$990,000 with probability $\frac{1}{2}$ and a *gain* of \$2,240,000 with probability $\frac{1}{2}$. Should she take the gamble? Explain. [Hint: $\sqrt{3,240,000} = 1800$.]

Problem 48 At some international airports, there are multiple passport control agents, with a single line serving several agents. (E.g., Heathrow airport.) When it is your turn, in such a one-line system, you go to the next available agent. At other international airports, each passport control agent has a separate line, and it is possible that if you choose badly, you will take longer to get through passport control than the average traveler; of course, if you choose well, you may end up taking much less time than average in such a multi-line system. (E.g., Frankfurt and Singapore airports.)

Suppose that everyone prefers to get through passport control as quickly as possible, and that no one has a superior advantage at guessing which lines will move faster. If the utility of a person's time can be represented in a similar manner as the utility of a person's money, explain why one system may be preferred to another depending upon a traveler's marginal utility of time. [Hint: you may want to think about risk preferences and diversification.]

Problem 49 True, False, Uncertain? [Explain your answer.] In the textbook model of costs, suppose that at the current level of output, the long-run average cost curve is decreasing, but the short-run average cost curve is increasing in output. *It follows that the long-run marginal cost curve must be below the short-run marginal cost curve at this level of output.*

Problem 50 A firm's cost function is $C(q) = q^2 + 16$. What is its marginal and average cost functions? Where is the minimum of average cost?

Problem 51 True, False or Uncertain? [Explain your answer.] A firm's long-run costs are currently \$100, of which \$50 represents factor payments to workers and \$50 represents payments to capital. *If wage rates increase by 50%, total long-run costs will go from \$100 to \$125.*

Problem 52 Suppose that a competitive firm can hire as many workers as it wants at the going wage of \$150 per day and can employ robots (capital) at a constant daily cost of \$300 each. At present, the firm is producing 200 cars per day with an input combination for which $MPP_K = 0.3$ and $MPP_L = 0.1$.

(a) What is this firm's marginal rate of technical substitution currently? Is this firm using an economically efficient input combination? If not, what should it do?

(b) In the short run (your robot level is fixed, but workers are variable), what are your marginal costs for an additional car at the present level of production?

Problem 53 True, False, or Uncertain? [Explain your answer!] Consider the textbook model of costs. *The long-run opportunity cost of producing a given level of output is always less than or equal to the short-run opportunity cost of producing that level of output.*

Problem 54 True, False, or Uncertain? [Explain your answer!] *If wages increase, then the short-run marginal cost of a competitive firm necessarily increases, while the long-run marginal cost could increase or decrease depending upon the particulars of the situation.*

Problem 55 True, False, or Uncertain. [Explain your answer!] All input prices rise by exactly 10 percent. *Both long-run and short-run marginal costs will increase by exactly 10 percent.*

Problem 56 True, False, Uncertain? [Explain your answer.] If average cost is constant for all levels of output, *then the marginal cost curve is flat and equal to the average cost curve.*

Problem 57 True, False, Uncertain? [Explain your answer.] For a given technology, whenever labor inputs double and capital inputs are held constant, output less than doubles. *This technology exhibits decreasing returns to scale.*

Problem 58 True, False, Uncertain? [Explain your answer.] Suppose that the prices of all short-run flexible inputs (e.g., labor) increase by 10 percent and short-run fixed inputs (e.g., capital) increase by 20 percent. *The short-run marginal cost of output will increase by more than 10 percent but less than 20 percent.*

Problem 59 Eric Clapton concert tickets cost \$45. You won a free ticket to see the Eric Clapton concert (which is not transferable and has no resale value). Bob Dylan is performing on the same night and is your next-best alternative activity. Both performers are only performing on this one night. Tickets to see Dylan cost \$40. You would be willing to pay up to \$50 to see Dylan if you don't go to the Clapton concert. Assume there are no other costs of seeing either performer. Based on this information, what is your opportunity cost of seeing Eric Clapton?

Problem 60 True, False, Uncertain? [Explain your answer.] Yesterday you were unexpectedly given a free ticket to an Coldplay concert scheduled for August 1. The market price of the ticket is \$75, but you can only sell it for \$50 if you don't use it yourself. Today, you

discover that U2 will be giving a concert that same evening (and only that evening). Tickets for the U2 concert are available at \$100. Had you not received the free Coldplay ticket, you would have bought a U2 ticket rather than a Coldplay ticket. *From what we are told of your preferences, it follows that if you are a rational utility maximizer, you should attend the U2 concert.*

Problem 61 Suppose that a competitive firm's cost of producing output q in the short-run is given by $C_{sr}(q) = 3q + q^2$. You may assume that there are no short-run capital expenditures, so variable cost and short-run cost are identical. Assume that for this firm all costs are variable in the short-run. The market price of the firm's product is \$9.

(a). What level of output will the firm produce?

(b). What is the firm's profit?

Problem 62 True, False, Uncertain? [Explain your answer.] *At the optimal point of production, if a firm's marginal revenue is above its average (opportunity) cost, then average (opportunity) cost must be increasing.*

Problem 63 In late July, a toy company receives an order for *up to* 75,000 toys to be delivered November 1. The contract calls for a price of \$20 each, payable on November 1. You may fulfill any portion of this order; you do not need to produce all 75,000 toys. Total monthly manufacturing costs, as a function of the number (in thousands) produced that month, are

$$C(q) = 80 + 0.5q^2$$

[You may assume that 80 is sunk if you like; your answer should not change.] Because this is such a large order, the company will be required to stockpile production until the November shipping date. However it will be required to pay storage costs of \$1 per toy per month until November 1. That is, a toy made in August will incur a storage cost of \$2 (September and October). (Worries about finance charges incurred between August 1 and November 1 should be put aside.) What is the optimal number of toys to be produced each month?

Problem 64 True, False or Uncertain [Explain your Answer!] *A firm should always produce at an output at which long-run average cost is minimized.*

Problem 65 You are in charge of a small aspirin manufacturing company. Each day you can make up to 1000 cases of generic aspirin. You can sell as many cases as you make for the market price of \$10 per case. Your weekly fixed operating costs are \$1000 (for rent and insurance). The cost to you for materials and supplies is \$4 per case; the cost of labor is \$5 per case, except on weekends, when it is \$10 per case (because of overtime labor costs). There are no sunk costs in producing aspirin.

- (a). How many cases of aspirin should you make in a week? What is your weekly economic profit?
- (b). Now suppose that in addition to making generic aspirin, you can also make a special “branded” version with the same costs as the generic version. You receive a special order for 1000 cases of branded aspirin per week for \$30 per case. Note that this is your only demand for your “branded” aspirin; your other production will be generic aspirin. How many cases of branded and generic aspirin should you produce each week?

Problem 66 A firm produces its first 30 units at a cost of \$ 10 per unit. Its next 20 units costs \$15 per unit. All additional units cost \$20 per unit. There are no fixed costs of production, so the marginal cost is equal to the average (opportunity) cost.

The firm can sell up to 40 units at a price of $p_A = \$12$ in market A , and up to 20 units at $p_B = \$17$ in market B . Nothing will be demanded at higher respective prices, and the same amount will be demand in each market at lower respective prices. Finally, assume that the firm’s costs only depend upon the total amount of output produced across both markets and that consumers cannot transport goods across market boundaries.

How much should the firm produce in each market?

Problem 67 Suppose that a firm can sell as much output as it likes at the market price of $p = 20$. Its opportunity cost of production in the short-run $60 + 2q^2$. (This implies that $MC_{sr}(q) = 4q$.) Its opportunity cost of production in the long-run is $90 + q^2$. (This implies that $MC_{lr}(q) = 2q$.)

- (a). What is the profit maximizing level of output in the short-run? Should the firm operate or shutdown? Explain.
- (b). What is the profit maximizing choice of output in the long-run? Should the firm operate or shutdown? Explain.

Problem 68 True, False, Uncertain? [Explain your answer.] Suppose that at a competitive firm’s current level of output, $p < MC$ and $p > AC$, where all costs are opportunity costs. *This firm could increase profit by increasing output.*

Problem 69 True, False, or Uncertain [Explain your Answer!] *A price low enough to cause a firm to shut down in the long run may not be low enough to cause it to shut down in the short run.*

Problem 70 Suppose that in a market with perfect competition, the market demand function is given by $Q^d = 1200 - 50p$. Every firm in the industry has a short-run *opportunity* cost of production given by

$$C_{sr}(q) = 2q^2.$$

- (a). Compute each firm's short-run supply curve.
- (b). If there are 600 firms, what is the short-run equilibrium price and quantity in this market?
- (c). If each firm's long-run *opportunity* cost function is given by

$$C_{lr}(q) = 2q^2 + 2 + q,$$

and there are an unlimited number of firms available with this cost function, what is the long-run equilibrium price in the market? How many firms will be producing?

Problem 71 The cheese business in Wisconsin is a competitive industry with an unlimited number of potential suppliers. All cheese manufacturers have the long-run opportunity cost function $C_{lr}(q) = q^2 + 9$. The demand for cheese manufactured in this town is given by

$$Q^d = 120 - p.$$

What is the long-run equilibrium number of firms in this industry?

Problem 72 Electricity is supplied competitively by two types of firms. The first type of firm has a very efficient (i.e., low-cost) form of power generation; this component of the market (which consists of dozens of competitive producers) will supply up to 10 Million megawatt hours (MMH) per day at a price of 1 per MMH. At any lower price, this market segment will supply nothing, but due to capacity constraints, a higher price than 1 will not produce greater supply than 10 MMH. [Hint: the supply curve for this segment is horizontal to 10 MMH, and then it becomes vertical.] The second segment of the market is less efficient, but similar in form. In aggregate, it will supply up to 5 MMH at a price of 2 per MMH. At any lower price, the inefficient segment will not supply any electricity; at a higher price than 2, this segment will only supply 5 MMH due to capacity concerns.

- (a). Graph the market's aggregate supply curve carefully labeling the points of interest on the curve. At a price of $p = \frac{1}{2}$, how much is supplied by the market? At a price of $p = \frac{3}{2}$, how much is supplied? At a price of $p = 3$, how much is supplied?
- (b). Suppose that the market demand curve for electricity is given by

$$P(Q) = 16 - Q.$$

What is the equilibrium price in this market? [Hint: graph the supply and demand curves rather than using algebra.] How much does each segment of suppliers produce at this equilibrium?

- (c). Suppose that government regulation forces the closure of all the inefficient plants due to pollution problems. What is the new equilibrium price and quantity?

Problem 73 Three firms are now manufacturing widgets for sale in a highly competitive market. Each firm has a capacity of seven units. Their costs are as follows:

Quantity	Firm A	Firm B	Firm C
1	1	1	1
2	3	4	2
3	6	8	4
4	10	13	7
5	15	19	11
6	21	26	16
7	28	34	22

The market demand for widgets equals: $25 - 2p$

- What is each firm's marginal cost and supply schedules?
- How many are produced? What is the equilibrium price?
- By which firms?

Problem 74 Suppose that in a market with perfect competition, the market demand function is given by $Q^d = 96 - p$ and the market supply function is given by $Q^s = 3p$. Every firm in the industry has a short-run opportunity cost of production (i.e., “variable” cost) of $C_{sr}^O(q) = 4q^2$, which has an associated marginal cost of $MC_{sr}(q) = 8q$.

- What is the short-run equilibrium price and quantity in this market?
- In the short-run equilibrium, how much does each firm produce? How many firms are in this industry?
- What is the total short-run producers surplus (i.e., profit) in this competitive market?
- Calculate the demand elasticity at the equilibrium price and quantity point.
- At this equilibrium point, if supply shifted inwards by a small amount, would market revenues go up or down?

Problem 75 Suppose that there are 100 firms operating smelters that each produce up to 1000 tons of aluminum. Ignoring the cost of the smelter (and just looking at short-run variable costs), the cost of producing aluminum (per ton) is \$250. Specifically, short-run opportunity cost is given by $C_{sr}^O(q) = 250q$ for any $q \leq 1000$. Thus, each firm is willing to supply 1000 tons of aluminum for any price greater than $p = \$250$, and they are unwilling to supply any output below a price of \$250.

- What do the (short-run) supply curves look like?
- If demand is given by

$$p = 1300 - 0.01 \cdot Q^d$$

what is the equilibrium price and quantity? What is each smelter's short-run economic profit (ignoring the cost of the smelter) at this price?

(c). Now suppose that demand unexpectedly shifts outward to

$$p = 1500 - 0.01 \cdot Q^d.$$

In the short run (given the number of smelters stays at 100), what is the new equilibrium price and economic profit of each firm?

(d). Suppose that the annual opportunity cost of a smelter is constant at \$50,000 per period. Thus, the long-run opportunity cost of smelting is $C_{lr}^O(q) = 250q + 50,000$ for $q > 0$ and $q \leq 1000$. (As a simplifying assumption, imagine that one can lease a smelter for this amount.) What market price of aluminum would be just sufficient to encourage a firm to buy/rent a smelter? Given this answer, what does the long-run industry supply curve for aluminum look like?

(e). Show/describe what happens to the industry short-run supply curves, the long-run supply curve and the market price when the market demand curve for aluminum unexpectedly shifts out from $p = 1300 - 0.01 \cdot Q^d$ to $p = 1500 - 0.01 \cdot Q^d$.

Problem 76 Suppose that in a market with perfect competition, the market demand function is given by $Q^d = 2200 - 100p$. Every firm in the industry has a short-run *opportunity* cost of production given by

$$C_{sr}^o(q) = 4q^2 + 2q + 4.$$

Note that the fixed cost of 4 is avoidable and not sunk.

(a). Compute each firm's short-run supply curve. [Be sure to indicate the range of prices for which the firms do not produce and to provide the formula for the quantity supplied in the price region in which they do produce.]

(b). If there are 800 firms, what is the short-run equilibrium price and quantity in this market?

(c). If each firm's long-run *opportunity* cost function is given by

$$C_{lr}^O(q) = 2q^2 + 8,$$

and there are an unlimited number of firms available with this cost function, what is the long-run equilibrium price in the market? How many firms will be producing?

Problem 77 Consider a constant-cost industry which is perfectly competitive market and comprised of homogeneous firms. Each firm has costs given by $C(q) = 10q$ (up to a capacity of 1 unit) and production is limited to 1 unit per firm. (Think of each firm representing one plant of 1-unit capacity; mergers are not allowed and technology requires 1-unit capacity, so firms never have any choice over scale.) In the long run, any firm can enter the industry for a per-period capital cost of \$75.

Timing: at the beginning of each period, each firm must make a planning decision (long-run decision) regarding whether or not to be in the market in the next period (i.e., whether or not to buy or sell capital). Each firm that has capital in place may also decide how much to produce in the current (e.g., short-run) period.

There are two possible demand curves in any period: low demand (given by $P(Q) = 110 - 2Q$) and high demand (given by $P(Q) = 310 - Q$). Each occurs with equal probability and these events are uncorrelated over time (i.e., high demand today tells us nothing about whether or not demand will be high tomorrow; high demand still arises with probability $\frac{1}{2}$). The realization of demand, however, will only be learned after capital decisions are sunk for that period.

(a). Suppose that initially there are $N = 100$ firms in the market. Calculate the short-run equilibrium price if demand is low, p_l , and the short-run equilibrium price when demand is high, p_h .

[Hint: it may be helpful to draw the industry's short-run supply function and the two different market demand curves in the same graph.]

(b). Given the current number of firms is $N = 100$, is each firm's expected short-run profit greater or less than the per-period cost of capital?

(c). How many firms will be in the industry if long-run expected economic profit is equal to zero? (I.e., what is the long-run equilibrium number of firms?) [Hint: Do not attempt to draw a long-run industry supply curve. Focus instead on the key condition of long-run equilibrium.]

Problem 78 In the competitive Yak butter industry in outer Bork, there are 10 identical, constant cost firms. Each has a long- and short-run marginal cost, MC, given by the equation:

$$MC_{sr} = MC_{lr} = 40 - 12q + q^2,$$

where q is the quantity produced by *the firm*. For each firm, the long run average cost is minimized at $q = 12$ units of Yak butter and short-run average cost is minimized at $q = 9$ units of Yak butter.

Market demand for the product in both the short and long run in this industry is given by

$$p = 160 - Q,$$

where Q is the total *market* output demanded and p is the price per unit.

(a). Explain why we know that the short-run average opportunity cost is $AC_{sr}^O = 13$ (i.e., $AVC=13$) when $q = 9$ and that the long-run average cost is $AC_{lr} = 40$ when $q = 12$.

(b). Using the the information above, is the industry in long-run equilibrium when the price is $p = 40$? Explain.

The following new fact is relevant for parts (c)-(d): As a result of a new free trade policy in Outer Bork, foreign sellers are now selling unlimited amounts of Yak butter at a price of 20 per unit.

(c). Are the domestic Yak butter firms going to shut down in the short-run? Explain.

(d). Analyze the long-run effects on the domestic Yak butter industry. Explain.

Problem 79 The industry demand for widgets is given by $Q = 240 - 10P$. Initially there are ten plants producing widgets. Each plant belongs to a different firm. (Indeed, there is a law restricting each firm to one plant.) Each has a cost function $16 + q^2$ where q is the number of widgets produced by each plant.

(a). Assuming initially that only these ten firm/plants may produce widgets, what will be the equilibrium price and quantity of widgets produced by each firm?

(b). If other firms may open a (single) plant and produce widgets, what will be the equilibrium price of widgets?

(c). How many widgets will each plant produce?

(d). How many firms will be producing widgets?

Problem 80 True, False, or Uncertain? [Explain your answer!] *In the short-run in a perfectly competitive industry with homogeneous firms, each firm will attempt to produce at the point at which its short-run average opportunity cost (i.e., $AC_{sr}^o(q) = AVC(q)$) is minimized.*

Problem 81 Consider a competitive industry which currently has 20 firms. There are 10 efficient firms and 10 inefficient firms. All firms have capacities of 1000 units; no output may be produced above capacity.

The efficient firms have $MC_{sr} = MC_{lr} = 100$, $AC_{sr}^O = AVC = 100$ and $AC_{lr}^O(q) = 100 + \frac{50,000}{q}$. Because the production efficiency of the low-cost firms is from learning and plant-specific process innovations, the technology is not transferable.

The inefficient firms have $MC_{sr} = MC_{lr} = 200$, $AC_{sr}^O = AVC = 200$ and $AC_{lr}^O(q) = 200 + \frac{50,000}{q}$. Hence, you can think of the inefficient firms having the same fixed costs of capital but different productivities from that capital.

Outside the industry there is a sufficiently large number of firms with the inefficient technology who represent the pool of potential entrants. Input costs are constant, regardless of industry size.

(a). Plot the short-run and long-run industry supply curves.

(b). If the industry is currently in long-run equilibrium and both types of firms supply output in equilibrium, what is the current market price?

(c). Suppose that the industry was in long-run equilibrium, but that demand suddenly shifted outward because of a *temporary* change in the economy. (Everyone knows the curve will shift back next period, however.)

1. What will happen to the short-run price (will it increase or decrease above the price in (b)) and the new long-run price (will it increase or decrease above the price in (b))?
2. Next period, who will enter or exit?
3. In the short-run, who will produce and how much will be produced?
4. Who will make short-run economic profits?

Problem 82 Suppose that long-run and short-run demand are identical (that is, both are represented by the same demand curve), but that short-run supply is more inelastic than long-run supply. Suppose also that all firms have identical cost structure and that the minimum short-run average opportunity cost function, $\min AC_{sr}^O(q)$ is greater than zero (i.e., the minimum AVC for each firm is greater than zero).

(a). Draw the three curves in one graph – demand and both short-run and long-run supply – when the market is in long-run equilibrium.

(b). Suppose that the demand curve shifts inward both unexpectedly and permanently. Describe the resulting changes of price over time – both the immediate change and the long-run change.

Problem 83 The industry demand for widgets is $Q = 600 - 10p$, where Q is the sum of all firms' production. Initially there are 40 plants producing widgets. Each plant belongs to a different firm. (A law restricts each firm to one plant.) Each plant has short-run economic costs equal to $C_{sr}^o(q) = 27 + 3q^2$ where q is the number of widgets produced by each plant. Short-run marginal costs for each plant are given by $MC_{sr}(q) = 6q$ and short-run average opportunity cost is $AC_{sr}^o(q) = 27/q + 3q$. These two curves intersect at $q = 3$.

(a). Assuming initially that only these 40 firms/plants may produce widgets and that the cost functions given above represent short-run economic cost, determine the short-run *industry* supply curve.

(b). (1) What is the equilibrium price in this market? (2) How many widgets will each firm produce? (3) What will be each firm's profits?

(c). Now assume that other firms may open a (single) plant and produce widgets if they wish in the long run. Also assume that the short-run and long-run cost functions are identical and all firms have access to this technology. Thus, long-run marginal and average opportunity costs are given by $MC_{lr}(q) = 6q$ and $AC_{lr}^o(q) = 27/q + 3q$. [Hence, the only distinction between the short-run and the long-run is that entry takes a period of time to occur.]

(d). (i) What is the equilibrium price in this market? (ii) How many widgets does each firm produce in the long run? (iii) How many firms are producing widgets? (iv) What is the profit of each firm?

Problem 84 True, False, or Uncertain. [Explain your answer!] *If all firms in an industry have the same cost conditions (i.e., identical technology, etc.), an increase in demand for the industry's product will not increase the price in the long run.*

Problem 85 True, False, or Uncertain? [Explain your answer!] *In the short-run in a perfectly competitive industry with homogeneous firms, each firm will attempt to produce at the point at which its short-run average opportunity cost (i.e., $AC_{sr}^o(q) = AVC(q)$) is minimized.*